Process Efficiency as a Barrier to Innovation

Online Appendix

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This appendix supplements "Process Efficiency as a Barrier to Innovation." Section, equation, table, figure, proposition, and footnote numbers without an appendix letter refer to the main text. Appendix A documents how subsidies and patents are distributed across firm sizes, B collects the production-side algebra, C the proofs, and D the dynamic choice of process efficiency. Appendix E constructs the breakthrough-patent measure, F reports robustness checks for the reduced-form results, G describes the standard errors, and H reports supplementary estimation results, including the restricted model, the crowding-out coefficient in the model, and the calibrated economy. Appendix I describes the computational algorithm, J the robustness of the quantitative results, and K an extension with heterogeneous firm types.

A. THE ALLOCATION OF SUBSIDIES AND PATENTS ACROSS FIRM SIZE

This appendix documents how innovation subsidies and patents are distributed across firm sizes in China. I group firms into the ten employment-size bins of the Economic Census classification of the National Bureau of Statistics. Since some industries are both larger and more subsidy-intensive than others, I compute all comparisons within 4-digit industry × year cells with at least 10 firms, which leaves 1,889 industry-years. I also reweight firms so that each size bin carries equal weight, which does not affect the subsidy yields. The two patterns below are unchanged, in results not reported here, when I instead reweight the stratified CTS sample to the bin shares of the National Economic Census or restrict the sample to 2016–2020.

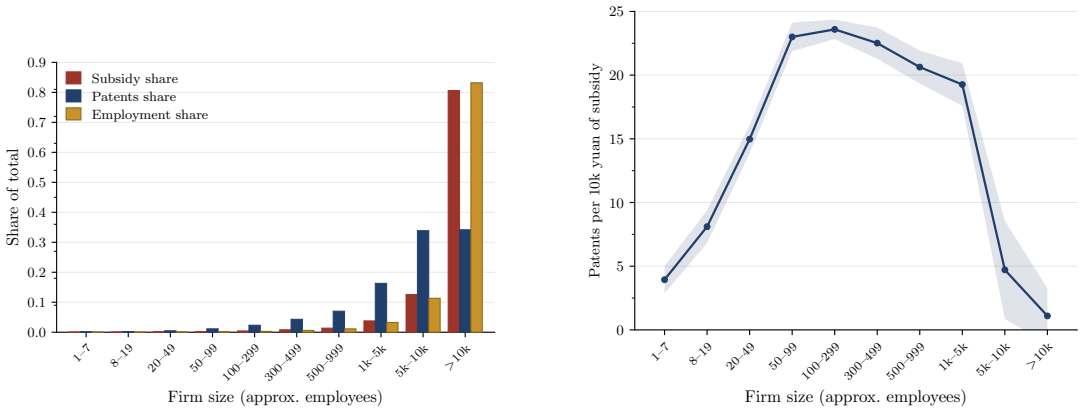
A.1. *Subsidy and patent shares*

Panel (a) of Figure A.1 shows the share of subsidies, patents, and employment in each size bin. Firms with more than 10,000 employees account for 81% of subsidies and 83% of employment but only 34% of patents. In the middle of the size distribution, patent shares exceed subsidy shares. Firms with 1,000–5,000 employees, for example, account for 16% of patents and 4% of subsidies. The sharpest break is between the 5,000–10,000 bin and the bin above 10,000 employees. The subsidy share jumps from 13% to 81% and the employment share from 11% to 83%, while the patent share remains

at about 34%. Hence subsidies track employment across firm sizes, and both diverge from patents.

A.2. *Patents per subsidy yuan*

I define the *subsidy yield* as the number of patents per 10,000 yuan of subsidy, computed among the firms that receive subsidies in each cell.[1] Panel (b) of Figure A.1 shows that the yield is hump-shaped in firm size. It rises from about 4 patents per 10,000 yuan at the smallest firms to a peak near 24 at 100–200 employees, stays high through the middle of the distribution, and falls toward zero above 3,000 employees. The 95% confidence bands at the peak and in the right tail do not overlap.



(a) Distribution of subsidies, patents, and employment across firm size.

(b) Patents per 10,000 yuan of subsidy by firm size.

FIGURE A.1
Subsidy allocation and yield across firm size

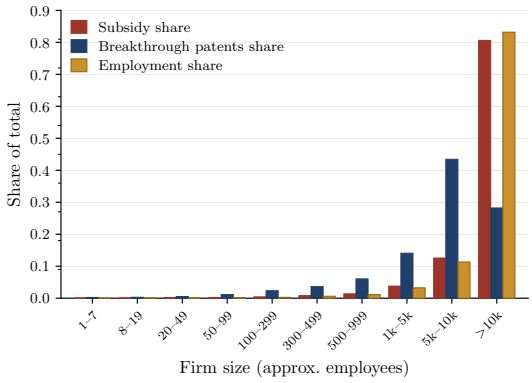
Notes: Panel (a) shows the shares of subsidies, patents, and employment accruing to each NBS employment-size bin when firms are reweighted so that every bin carries equal weight. Panel (b) shows patents per 10,000 yuan of subsidy among recipients, with the 95% confidence band computed from between-cell variation; the yield profile is invariant to the reweighting. Yields are winsorized at the 99th percentile. Bin-level statistics require at least three subsidized firms per cell and five contributing cells per bin.

A.3. *Robustness to strategic patenting*

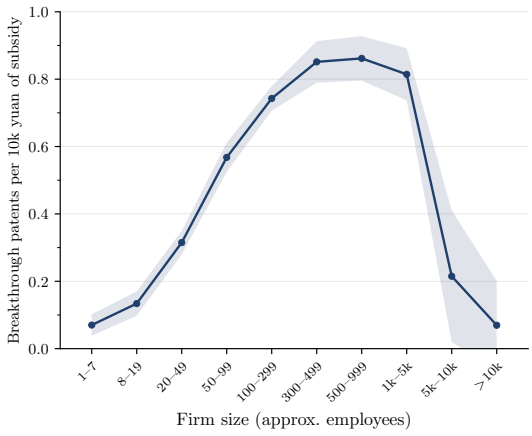
Chinese firms have well-documented incentives to file low-quality patents in order to qualify for High-Technology Enterprise status, tax preferences, and procurement eligibility. If small and mid-sized firms file for these reasons more often than large firms, the patterns above could reflect filing behavior rather than innovation. I therefore reconstruct both panels of Figure A.1 using only breakthrough patents, the top 10%

1. The yield is defined only for recipients. If the small firms that receive subsidies are more capable than the typical small firm, the hump could partly reflect selection rather than the return to a marginal yuan of subsidy. The regressions in Section 3.2 do not have this problem, since they use all firms and exploit variation in subsidy concentration within industries.

of patents by the inventiveness measure of Appendix E. Figure A.2 shows that both patterns remain. Firms with more than 10,000 employees account for 81% of subsidies but only 28% of breakthrough patents, which are concentrated in the 1,000–5,000 and 5,000–10,000 bins (14% and 43%). Breakthrough patents per yuan of subsidy have the same hump shape, with the highest yields at 200–2,500 employees and yields close to zero above 5,000. Hence the mismatch between where subsidies go and where patents come from is not driven by low-quality filings.



(a) Distribution of subsidies, breakthrough patents, and employment.



(b) Breakthrough patents per 10,000 yuan of subsidy.

FIGURE A.2
Subsidy allocation and breakthrough patent yields across firm size

Notes: Breakthrough patents are the top decile of year-demeaned log inventiveness (Appendix E). In panel (a), each bar shows the share of subsidies, breakthrough patents, or employment accounted for by firms in that size bin when the bins are reweighted to equal firm weight. Panel (b) reports yields among subsidy recipients, winsorized at the 99th percentile before aggregation, with the shaded 95% confidence band. All statistics are computed within 4-digit industry × year cells.

B. PRODUCTION-SIDE ALGEBRA

This appendix derives the profit coefficient $\tilde{p}$ of Section 2.1. Given prices $\{p_j\}$, the final-good producer solves

$$\max_{\{k_j\}} \; L^{\beta} \int_{j\in\mathcal{N}} z_j^{\beta} k_j^{1-\beta}\, dj - \int_{j\in\mathcal{N}} p_j k_j \, dj,$$

with first-order condition

$$k_j = (1-\beta)^{1/\beta}\, L\, z_j\, p_j^{-1/\beta}.$$

Demand has constant elasticity $1/\beta$ and every line has marginal cost $w_u/\bar{q}$, so the leader charges $p_j = w_u/[(1-\beta)\bar{q}]$ on every active line. Substituting,

$$k_j = (1-\beta)^{2/\beta}\, L\, z_j\, \bar{q}^{\,1/\beta} w_u^{-1/\beta}, \qquad \Pi_j = \frac{\beta}{1-\beta}\,\frac{w_u}{\bar{q}}\, k_j = \beta\,(1-\beta)^{\frac{2}{\beta}-1} L\, z_j\, \bar{q}^{\,\frac{1}{\beta}-1} w_u^{\,1-\frac{1}{\beta}}.$$

Since $z_j^{\beta} k_j^{1-\beta} \propto z_j$, aggregate output is $Y = (1-\beta)^{2(1-\beta)/\beta} L\, \bar{q}^{\,(1-\beta)/\beta} w_u^{-(1-\beta)/\beta} Z$, where $Z \equiv \int_{j\in\mathcal{N}} z_j\, dj$. The first-order condition for assembly labor, $w_u = \beta Y/L$, then gives the wage

$$w_u = \beta^{\beta}(1-\beta)^{2(1-\beta)}\, \bar{q}^{\,1-\beta}\, Z^{\beta}.$$

Substituting back, the powers of $1-\beta$ collect to $\frac{2-2(1-\beta)^2}{\beta} - 1 = 3-2\beta$ and those of $\bar{q}$ to $\frac{1-(1-\beta)^2}{\beta} - 1 = 1-\beta$, so that

$$\Pi_j = \beta^{\beta}(1-\beta)^{3-2\beta}\, L\, z_j\, \bar{q}^{\,1-\beta}\, Z^{\beta-1}. \tag{B.1}$$

Line profit (B.1) is linear in $z_j = q_j(1+\psi_f)$. Hence normalized line profit is

$$\frac{\Pi_j}{\bar{q}} = \tilde{p}\, x_j\, (1+\psi_f), \qquad \tilde{p} = \beta^{\beta}(1-\beta)^{3-2\beta}\, L\, (Z/\bar{q})^{\beta-1},$$

and summing over the lines of a firm gives operating profit $\tilde{p}\,(1+\psi_f)\,Q_f$. Since $\bar{q}$ is the average quality of active lines, $Z = \bar{q}\int_{\mathcal{N}} x_j(1+\psi_{f(j)})\, dj = \bar{q}\,\Phi_A\bar{\Psi}$, where

$$\bar{\Psi} \equiv \int x\,(1+\psi')\, dS(x,\psi')$$

is the quality-weighted average of $1+\psi$ across active lines. Therefore

$$\tilde{p} = \beta^{\beta}(1-\beta)^{3-2\beta}\, L\, (\Phi_A\bar{\Psi})^{\beta-1}, \tag{B.2}$$

Since $\beta < 1$, higher average process efficiency lowers $\tilde{p}$.

Two further results are used in the main text. First, output is $Y = \beta^{\beta-1}(1-\beta)^{2(1-\beta)} L\, \bar{q}\, (\Phi_A\bar{\Psi})^{\beta}$, so Y and $\bar{q}$ grow at the same rate on the balanced growth path. Second, unskilled labor in intermediate production is $\int_{\mathcal{N}} k_j/\bar{q}\, dj = (1-\beta)^2 L/\beta$, a constant multiple of assembly labor, as footnote 2 states.

C. PROOFS

This appendix contains the proofs of Propositions 1–4, the derivation of the growth rate (19), and the general-equilibrium effects of a subsidy discussed in Section 2.6. Notation follows Section 2. In addition, define

$$\mathcal{L}f(p) \equiv \sum_j [f(p) - f(p \ominus x_j)],$$

so that $\tau(\psi(p))\,\mathcal{L}\hat{V}(p)$ is the expected loss from creative destruction in (14).

Two lemmas

LEMMA C.1. *Consider a defender with process efficiency ψ that faces challengers i with attempt intensities $a_i \geq 0$ and process efficiencies $\psi_i \geq 0$, so that each of its lines is displaced at rate*

$$T(\psi) = \sum_i a_i \,\Lambda\big((1+\lambda)(1+\psi_i) - (1+\psi)\big).$$

Then T is decreasing in ψ, and $T(\psi) \to 0$ as $\psi \to \infty$ if $\sum_i a_i < \infty$. In the step-function limit $\kappa \to \infty$, a challenger with $\psi_i = 0$ contributes zero to T if $\psi > \lambda$.

Proof. The advantage of every challenger is decreasing in ψ and Λ is increasing, so every term of T is decreasing in ψ, and hence so is T. As $\psi \to \infty$, every term converges to zero. Since the i-th term is bounded by a_i and $\sum_i a_i < \infty$, dominated convergence gives $T(\psi) \to 0$. The same argument applies when the sum is an integral over challengers. Finally, a challenger with $\psi_i = 0$ has advantage $\lambda - \psi$, and as $\kappa \to \infty$, $\Lambda(\lambda - \psi) \to \mathbf{1}\{\psi < \lambda\}$ for $\psi \neq \lambda$, which is zero for $\psi > \lambda$. □

LEMMA C.2. *Suppose that the subsidy rate $\sigma(p)$ is weakly increasing in size, which includes the case without subsidies. Then $\hat{V}(p \oplus y) > \hat{V}(p)$ for every portfolio p and every $y > 0$.*

Proof. Denote by A a firm with portfolio $p \oplus y$ and by B a firm with portfolio p, and let A choose the attempt rate of B in every period. Put the two firms on a common probability space: their attempts arrive at the same times and target the same lines, and each attempt succeeds or fails according to the same uniform draw. Both portfolios are then random, but they can be compared sample path by sample path. Denote by $p_A(t)$ and $p_B(t)$ the portfolios in period t.

We show that, along every sample path, the lines of $p_B(t)$ are a subset of the lines of $p_A(t)$ in every period. This holds in period 0. Suppose that it holds in period t. Then $Q(p_A(t)) \geq Q(p_B(t))$ and, since σ is weakly increasing in size, $\sigma(p_A(t)) \geq \sigma(p_B(t))$, so $\psi(p_A(t)) \geq \psi(p_B(t))$. Since the success probability (11) is increasing in process efficiency, A succeeds whenever B does. By Lemma C.1, A also loses each common line at a weakly lower rate, so the draws can be chosen such that it loses such a line only if B does. Since the two firms exit together and the relative qualities of their lines fall

at the same rate g, the property holds after period t as well.

Hence $Q(p_A(t)) \geq Q(p_B(t))$ along every sample path. In every period, A therefore earns at least the flow profit of B, strictly more while it holds a line that B does not, at an R&D cost $(1 - \sigma(p_A(t)))\, w_s h_X$ that is weakly lower. Taking expectations, the value of A from copying the attempt rate of B strictly exceeds $\hat{V}(p)$. Since $\hat{V}(p \oplus y)$ is the maximum value over all policies of a firm with portfolio $p \oplus y$, we have $\hat{V}(p \oplus y) > \hat{V}(p)$. □

Proof of Proposition 1 (Process efficiency insulates large incumbents).

Without subsidies, $\psi(p) = \psi^*(Q(p))$, and the displacement rate (12) is a special case of T in Lemma C.1. There are two groups of challengers: entrants, with intensity X_e and zero process efficiency, and incumbents with portfolio p'', with intensity $X^*(p'')\, dM(p'')$ and process efficiency $\psi(p'')$. Their total intensity is the aggregate attempt rate $\bar{\tau} < \infty$.

Part (a). By (5), $\psi^*(Q)$ is increasing in Q and $\psi^*(Q) \to \infty$ as $Q \to \infty$. Lemma C.1 therefore implies that $\tau(\psi^*(Q))$ is decreasing in Q and converges to zero.

Part (b). For $Q > \bar{Q}_e$, (10) gives $\psi^*(Q) > \lambda$, so by Lemma C.1 no entrant displaces the firm in the step-function limit.[2] An incumbent with process efficiency ψ'' displaces the firm if and only if $(1 + \lambda)(1 + \psi'') > 1 + \psi^*(Q)$, that is,

$$\psi'' > \frac{1 + \psi^*(Q)}{1 + \lambda} - 1.$$

The right-hand side is increasing in Q, so the set of incumbents that can displace the firm shrinks as Q rises. □

Proof of Proposition 2 (Larger firms devote more R&D labor to process efficiency).

Without subsidies, $h_\psi(p) = c_\psi[\psi^*(Q)]^\omega \propto Q^\eta$ by (6), so it suffices to show that h_X grows more slowly than Q^η. The proof strategy is to show that firm value grows like Q^η. The gain from an attempt then grows like $Q^{\eta-1}$ and, by (15), product R&D labor grows like $Q^{(\eta-1)/\gamma}$.

To this end, conjecture that (14) has a solution with $\hat{V}(p) = V_0\, Q(p)^b\, (1 + o(1))$ as $Q(p) \to \infty$, for some $b \geq 1$ and $V_0 > 0$, and with increments and derivatives of the corresponding order. We determine b by computing the order of each term in (14).

Gain from an attempt. For $b \geq 1$ and $y \geq 0$,

$$(Q + y)^b - Q^b \leq b\, (Q + y)^{b-1} y \leq b\, 2^{b-1} (Q^{b-1} y + y^b).$$

The added line has relative quality $y = 1 + \lambda$ after an attempt on an inactive line and $y = (1 + \lambda)x$ after a capture. Since $\int x\, dS = 1$, we have $G(p) = O(Q^{b-1})$ whenever $\int x^b\, dS < \infty$. By condition (vi) of Definition 1, this holds at $b = \eta$. By (15),

$$h_X(p) = \left[\frac{(1 - \gamma)\, \theta\, G(p)}{w_s}\right]^{1/\gamma} = O\big(Q^{(b-1)/\gamma}\big),$$

2. At $Q = \bar{Q}_e$ the advantage of an entrant is zero, and $\Lambda(0) = 1/2$ for every κ.

and the R&D surplus at the optimum, $X^*G - w_s h_X = \frac{\gamma}{1-\gamma}\, w_s h_X$, is of the same order.

Loss of lines. For $b \geq 1$, $Q^b - (Q - x_j)^b \leq b\, Q^{b-1} x_j$, so

$$0 \leq \mathcal{L}\hat{V}(p) \leq b\, V_0\, Q^{b-1} \sum_j x_j\, (1 + o(1)) = b\, V_0\, Q^b\, (1 + o(1))$$

for any composition of p. Since $\tau(\psi^*(Q)) \to 0$ by Proposition 1, the loss term in (14) is $o(Q^b)$.

Decline in relative quality. We have $\sum_j x_j\, \partial \hat{V}/\partial x_j = b\, V_0\, Q^b\, (1 + o(1))$.

Matching orders. The left-hand side of (14) is of order Q^b. On the right-hand side, the flow profit is of order Q^η, the R&D surplus is of order $Q^{(b-1)/\gamma}$, and the remaining terms are of order at most Q^b. Hence

$$b = \max\left\{\eta,\ \frac{b-1}{\gamma}\right\},$$

which has the two roots $b = \eta$ and $b = 1/(1-\gamma)$. Assumption 2 is equivalent to $\eta < 1/(1-\gamma)$, since $\eta = \omega/(\omega - 1)$ is decreasing in ω and equals $1/(1-\gamma)$ at $\omega = 1/\gamma$.

First consider $b = 1/(1-\gamma)$. Then $h_X = O(Q^{1/(1-\gamma)})$ grows faster than the flow profit, so the flow payoff $\pi - w_s h_X$ diverges to $-\infty$. Since $\hat{V} > 0$, the value must then come entirely from the limit term, which violates the transversality condition $\lim_{T\to\infty} e^{-\hat{r}T}\, \mathbb{E}[\hat{V}(p_T)] = 0$. We therefore rule out this root.

Next consider $b = \eta$. Assumption 2 implies $(\eta - 1)/\gamma < \eta$, so the R&D surplus is of lower order than the flow profit. Matching the terms of order Q^η gives $\hat{r}\, V_0 = \hat{D} - g\, \eta\, V_0$, that is,

$$V_0 = \frac{\hat{D}}{\hat{r} + \eta\, g}.$$

The constant V_0 does not depend on the composition of p, as the conjecture requires, since the only term in which the composition enters, the loss term, is of lower order.

Hence $h_X(p) = O(Q^{(\eta-1)/\gamma}) = O(Q^{1/[\gamma(\omega-1)]})$, while $h_\psi(p) \propto Q^\eta$. Comparing exponents,

$$\eta > \frac{1}{\gamma(\omega-1)} \iff \frac{\omega}{\omega-1} > \frac{1}{\gamma(\omega-1)} \iff \omega > \frac{1}{\gamma}.$$

Under Assumption 2, therefore, $h_X(p)/h_\psi(p) \to 0$ and $\chi(p) \to 1$ for any composition of p. □

Remark. The argument does not show that χ is monotone in Q, since h_X/h_ψ contains lower-order terms that need not vanish monotonically.

On condition (vi). To see when condition (vi) holds, consider the relative quality of a single line over time: while the line is active, its relative quality falls at rate g as average quality grows, and it jumps by the factor $1 + \lambda$ whenever the line is improved. When its owner exits, at rate φ, the line becomes inactive, and it later re-enters the economy at quality $(1 + \lambda)\bar{q}$, which is close to the economy-wide mean. Exit thus pulls every line back toward the mean, and this keeps the distribution of relative quality

from spreading out forever. Since exit is rare relative to improvement, however, the right tail is thick. Suppose, for example, that every active line is displaced, and hence improved, at a common rate ν. The moment of order k of relative quality is then finite if and only if

$$\nu\big[(1+\lambda)^k - 1\big] - g\,k < \varphi,$$

where $g = \lambda(\nu + \varphi)$ by (19). At the calibrated values $\lambda = 1.08$, $\varphi = 0.004$, and $g = 0.023$, the largest such k is about 1.7. This is above $\eta \approx 1.44$, so condition (vi) holds, though not by a wide margin. The tail is thinner still if high-quality lines tend to be held by firms with high process efficiency, since those lines are displaced, and hence improved, less often.

Proof of Proposition 3 (Subsidies to large firms fund process efficiency).

The proof uses the idea that a subsidy scales each type of R&D by a fixed factor, which means that the firm puts the extra labor in the activity to which it already allocates most of its R&D. Consider a marginal subsidy to a firm with portfolio p, which lowers its effective skilled wage to $(1-\sigma)w_s$. The firm is atomistic, so it takes w_s, its success probabilities, and $\hat{V}(\cdot)$ as given.

Process efficiency solves (4) at the subsidized wage, so $\psi^*(Q;\sigma) = (1-\sigma)^{-1/(\omega-1)}\psi^*(Q)$ and

$$h_\psi(p;\sigma) = c_\psi\big[\psi^*(Q;\sigma)\big]^\omega = (1-\sigma)^{-\eta}\,h_\psi(p).$$

With w_s, the success probabilities, and $\hat{V}$ fixed, the expected gain $G(p)$ is fixed, and (15) gives $h_X(p;\sigma) = (1-\sigma)^{-1/\gamma}h_X(p)$. Differentiating each at $\sigma = 0$,

$$\frac{\partial h_\psi}{\partial \sigma} = \eta\,h_\psi(p), \qquad \frac{\partial h_X}{\partial \sigma} = \frac{1}{\gamma}\,h_X(p),$$

so the share of the additional R&D labor that goes to process efficiency is

$$\varsigma(p) = \frac{\partial h_\psi/\partial\sigma}{\partial h_\psi/\partial\sigma + \partial h_X/\partial\sigma} = \frac{\gamma\eta\,\chi(p)}{\gamma\eta\,\chi(p) + 1 - \chi(p)}.$$

Differentiating, $\partial\varsigma/\partial\chi = \gamma\eta/[\gamma\eta\chi + 1 - \chi]^2 > 0$, so ς is strictly increasing in χ. By Proposition 2, $\chi(p) \to 1$ under Assumption 2, so $\varsigma(p) \to 1$. □

Proof of Proposition 4 (Subsidies to large firms discourage innovation by smaller firms).

Consider a subsidy to large firms, and let p be the portfolio of a smaller firm, whose subsidy rate therefore does not change. By (5), with $(1-\sigma(p))\,w_s$ in place of w_s, the subsidy raises the process efficiency of every subsidized firm and leaves that of every other firm unchanged.

Part (a). Consider the success probability (11). Since the distribution of firms is fixed, so are Φ_A and the weights of S, and the term for inactive lines is unchanged. In

the term for active lines, the advantage $A(\psi(p), \psi')$ of the smaller firm falls on every line held by a subsidized firm and is unchanged on every other line. Since Λ is increasing, $B(\psi(p))$ falls, strictly if subsidized firms hold a positive share of active lines and $\kappa < \infty$. The same argument with $\psi(p)$ replaced by zero shows that $B_e = B(0)$ falls.

Part (b). By (13), the term for inactive lines is unchanged. In each capture term, the success probability $\Lambda(A(\psi(p), \psi'))$ falls or is unchanged by part (a), while the gain $\hat{V}(p \oplus (1+\lambda)x) - \hat{V}(p)$ is fixed. Since σ remains weakly increasing in size, this gain is positive by Lemma C.2. Hence $G(p)$ falls, and since w_s is fixed, (15) implies that $X^*(p)$ falls. For an entrant, each gain is the value $\hat{V}(\{(1+\lambda)x\})$ of a one-line portfolio, which is positive by Lemma C.2 since $\hat{V}(\emptyset) = 0$. Hence $G(\emptyset)$ falls, and by (16) so does X_e. □

Derivation of the growth rate (19)

Let $T \equiv \int_{\mathcal{N}} q_j \, dj = \bar{q}\,\Phi_A$ be the aggregate quality of active lines. On the balanced growth path Φ_A is constant, so $g = \dot{T}/T$. T rises when active lines are improved or inactive lines re-enter the economy, and it falls when lines exit. An active line held by a firm with process efficiency ψ' is displaced, and hence improved by the factor $1+\lambda$, at rate $\tau(\psi')$, which adds $\lambda\,\bar{q}\,\Phi_A \int x\,\tau(\psi')\,dS(x,\psi')$ per unit of time. Inactive lines receive attempts at rate $\bar{\tau}$, and each re-enters the economy at $(1+\lambda)\bar{q}$ after its first attempt, which adds $(1+\lambda)\,\bar{q}\,\bar{\tau}\,\Phi_I$. Lines exit at rate φ regardless of their quality, which removes $\varphi\,\bar{q}\,\Phi_A \int x\,dS = \varphi\,\bar{q}\,\Phi_A$. Stationarity of Φ_A requires $\bar{\tau}\,\Phi_I = \varphi\,\Phi_A$. Hence

$$\dot{T} = \bar{q}\,\Phi_A\Big[\lambda \int x\,\tau(\psi')\,dS(x,\psi') + (1+\lambda)\varphi - \varphi\Big],$$

and dividing by T gives (19).

Growth effects of a subsidy to large firms

Direct innovation (positive). The subsidy lowers the effective wage at subsidized firms and raises $X^*(p)$ through (15). By (12), this raises the displacement rate of every line they attack, hence g in (19) rises.

Blocking (negative). The subsidy raises ψ at subsidized firms. By Lemma C.1, the displacement rate of their lines falls, which lowers the quality-weighted displacement rate in (19).

Crowding out (negative). By Proposition 3, the additional demand for labor at large subsidized firms is concentrated in process R&D. Aggregate demand for skilled labor rises, and with supply fixed at L^S the wage w_s must rise to clear (18). The higher wage lowers $X^*(p)$ at unsubsidized firms and lowers X_e, which reduces displacement rates and growth.

D. STATIC VS. DYNAMIC CHOICE OF PROCESS EFFICIENCY

In the main text, process efficiency solves the static problem (4), so a firm chooses ψ only to raise its current profit. A higher ψ, however, also raises the firm's chances of

capturing lines and lowers the rate at which it loses them. In this appendix I let the firm take these two effects into account and show that the static formulation understates process efficiency for every portfolio.[3]

D.1. *Dynamic value function*

Let $\hat{V}(p;\psi)$ denote the value, normalized by $\bar{q}$, of a firm with portfolio p that maintains process efficiency ψ at the current instant. I keep continuation values at the value function $\hat{V}$ of (14). The question is then how a firm would set ψ today if it took into account that a higher ψ helps it acquire new lines and block potential captures. The value of such a firm is

$$\begin{aligned}\hat{r}\,\hat{V}(p;\psi) = \tilde{p}\,(1+\psi)\,Q(p) - w_s c_\psi \psi^\omega - w_s h_X(p) \\ + X^*(p)\,\mathcal{G}_\psi \hat{V}(p) - \tau(\psi)\,\mathcal{L}\hat{V}(p) - g\sum_j x_j\,\frac{\partial \hat{V}(p)}{\partial x_j},\end{aligned} \tag{D.1}$$

where $\mathcal{G}_\psi$ is the operator (13) with the firm's own process efficiency set to ψ, and $\mathcal{L}$ is the loss operator of Appendix C. Besides current profit, the firm's own ψ enters (D.1) through the expected gain from an attempt,

$$\begin{aligned}\mathcal{G}_\psi \hat{V}(p) = \Phi_I\big[\hat{V}(p \oplus (1+\lambda)) - \hat{V}(p)\big] \\ + \Phi_A \int \Lambda\big((1+\lambda)(1+\psi) - (1+\psi')\big)\big[\hat{V}(p \oplus (1+\lambda)x) - \hat{V}(p)\big]\, dS(x,\psi'),\end{aligned}$$

which rises with ψ because a higher ψ raises the firm's chance of winning every contest and each gain is positive by Lemma C.2, and through the displacement rate $\tau(\psi)$ in (12), which falls with ψ by Lemma C.1.

D.2. *Dynamic first-order condition*

Differentiating (D.1) with respect to ψ and holding continuation values fixed, we obtain the first-order condition

$$\underbrace{\omega\, w_s\, c_\psi\, \psi^{\omega-1}}_{L(\psi)} = \underbrace{\tilde{p}\,Q(p) + \underbrace{X^*(p)\,\frac{\partial \mathcal{G}_\psi \hat{V}(p)}{\partial \psi}}_{\text{Attack term } \mathcal{A}(p;\psi)} + \underbrace{|\tau'(\psi)|\,\mathcal{L}\hat{V}(p)}_{\text{Defense term } \mathcal{D}(p;\psi)}}_{R(p;\psi)}. \tag{D.2}$$

By the envelope theorem, the response of X^* to ψ does not enter (D.2). The left-hand side is the marginal cost of process efficiency. On the right, $\tilde{p}\,Q(p)$ is the static benefit, the attack term is the value of winning more contests, and the defense term is the value of losing lines less often. Setting $\mathcal{A} = \mathcal{D} = 0$ gives back (5).

3. Process efficiency is still chosen at every instant, so ψ never becomes a state variable. A formulation in which today's ψ persists and constrains tomorrow's is a different problem, which I do not solve here.

PROPOSITION D.1 (The static formulation is a lower bound). *Equation* (D.2) *has at least one solution, and every solution satisfies* $\psi_{\text{dyn}}(p) > \psi^*(Q(p))$ *for every nonempty portfolio* p.

Proof. For existence, note that $L(0) = 0$ while $R(p;0) \geq \tilde{p}\,Q(p) > 0$. The slope of the logistic function is bounded, $\Lambda' \leq \kappa/4$, so

$$\frac{\partial \mathcal{G}_\psi \hat{V}(p)}{\partial \psi} \leq \frac{(1+\lambda)\kappa}{4} \int [\hat{V}(p \oplus (1+\lambda)x) - \hat{V}(p)]\, dS(x, \psi') \qquad \text{and} \qquad |\tau'(\psi)| \leq \frac{\bar{\tau}\,\kappa}{4}.$$

The integral is finite by condition (vi) of Definition 1, and $\mathcal{L}\hat{V}(p)$ is finite, so $R(p;\cdot)$ is bounded above. Since L is continuous and unbounded and $R(p;\cdot)$ is continuous and bounded, $L - R$ changes sign, and a root exists by the intermediate value theorem.

We next show that every root exceeds $\psi^*(Q(p))$. Both correction terms are strictly positive at every ψ. In the attack term, $X^*(p) > 0$ because $G(p) > 0$ by Lemma C.2, $\Lambda' > 0$, and each increment is positive by the same lemma. In the defense term, $|\tau'(\psi)| > 0$, and $\mathcal{L}\hat{V}(p) > 0$ for a nonempty portfolio by the lemma. Hence $R(p;\psi) > \tilde{p}\,Q(p)$ for every ψ. Let ψ_{dyn} be any root. Then

$$L(\psi_{\text{dyn}}) = R(p;\psi_{\text{dyn}}) > \tilde{p}\,Q(p) = L(\psi^*(Q(p))),$$

where the last equality is the static first-order condition. For $\omega > 1$, L is strictly increasing on $\psi > 0$, so $\psi_{\text{dyn}} > \psi^*(Q(p))$. □

Since Λ' first rises and then falls, $R(p;\cdot)$ need not be monotone, so there may be several roots, all of which exceed the static solution.

Both corrections are small for a firm that wins or loses almost every contest with certainty, since Λ' is then close to zero. The attack motive applies to entrants as well, who would choose $\psi > 0$ to raise B_e. This works against blocking, so the implications below concern incumbents, and the net effect on entry is ambiguous.

D.3. *Implications for the results of the paper*

1. **Process R&D is understated.** Process R&D labor $h_\psi = c_\psi \psi^\omega$ is increasing in ψ, which is larger under the dynamic formulation for every portfolio.

2. **Blocking is understated.** The advantage $A(\psi'', \psi) = (1+\lambda)(1+\psi'') - (1+\psi)$ falls with the defender's ψ. Holding challengers' process efficiency fixed, a higher ψ at incumbents therefore raises the probability $1 - \Lambda(A(\psi'', \psi))$ that an attempt fails.

3. **The welfare gains from reallocation are understated.** To the extent that the gains in Section 5 operate through these two channels, the static formulation understates them.

E. BREAKTHROUGH PATENT CONSTRUCTION

This appendix describes the breakthrough-patent measure used in Appendix A and Section 3.2. I adopt the text-based inventiveness score of Kelly et al. (2021), which is based on the content of each patent's claims rather than its citations.

The score asks whether the text of a patent resembles later patents more than earlier ones.

I construct the score in four steps. First, I tokenize the claims of each patent with the Jieba segmenter, the standard tool for Chinese word segmentation. Second, I filter the vocabulary. I keep the 435,005 tokens that appear in at least 10 patents and ask the DeepSeek large language model to classify each of them as technical or non-technical. The 199,782 technical tokens form the final vocabulary. Without this filter, common characters and grammatical particles would dominate the token counts. Third, I represent each patent by its term frequency–inverse document frequency (TF-IDF) vector over this vocabulary, namely the count of each token in the claims weighted by the log of the inverse share of patents that contain it. I compute the weights from patents filed in the ten years before the focal patent, so that they use only information available at the filing date.

Fourth, for each patent i filed in year t_i, I compute the average cosine similarity between its vector and those of patents filed in the ten years before t_i, and the same average for the ten years after t_i. Both averages are taken within the patent's IPC section, so that similarity reflects technological proximity rather than accidental overlap between unrelated fields. The inventiveness score is the ratio of the two,

$$\mathrm{inv}(i) \;=\; \frac{\mathrm{sim}_{\mathrm{fwd}}(i)}{\mathrm{sim}_{\mathrm{back}}(i)},$$

so a patent with $\mathrm{inv}(i) > 1$ is closer to later filings than to earlier ones. The measure differs from that of Kelly et al. (2021) in three ways: the Chinese tokenization, the filtering of the vocabulary by a language model, and the restriction of comparisons to the same IPC section.

To obtain a breakthrough indicator, I take the log of $\mathrm{inv}(i)$ and subtract its mean in each year. The demeaning removes an upward trend in raw inventiveness that arises mechanically as the pool of comparison patents grows. I then classify the top 10% of the demeaned distribution as breakthrough patents.

F. REDUCED-FORM ROBUSTNESS

This appendix checks the results of Table 1 in five ways: alternative cutoffs for subsidy concentration, breakthrough patents as the dependent variable, two-way clustering, restrictions on the sample, and a pre-trend test of the Bartik instrument.

F.1. *Alternative cutoffs*

The concentration measure Conc^p_{jt} defines large firms as those above the 90th percentile of employment in the cell. Table F.1 reruns the column 3 specification with $p \in \{50, 75, 95, 99\}$. The small-firm coefficient is negative and significant at every cutoff and under both estimators. At $p = 50$ and $p = 75$ its magnitude is close to the baseline. At $p = 95$ and $p = 99$ both the small- and the large-firm coefficients rise in absolute value, as the group of large firms becomes small, which makes its subsidy share volatile.

TABLE F.1
Alternative concentration cutoffs for the column 3 specification

	Concentration cutoff p				
	50	75	90	95	99
Panel A: OLS					
$\text{Conc}^p \times$ Small	-0.889^{***}	-0.738^{***}	-0.682^{***}	-0.713^{***}	-0.951^{***}
	(0.319)	(0.172)	(0.171)	(0.196)	(0.284)
$\text{Conc}^p \times$ Large	0.395	0.063	0.240	0.375^{**}	0.717^{***}
	(0.319)	(0.213)	(0.189)	(0.183)	(0.214)
Panel B: Bartik IV					
$\text{Conc}^p \times$ Small	-1.952^{***}	-0.897^{***}	-1.559^{***}	-2.582^{***}	-5.917^{***}
	(0.672)	(0.274)	(0.343)	(0.524)	(1.262)
$\text{Conc}^p \times$ Large	0.247	−0.010	0.283	0.899^{***}	3.897^{***}
	(0.433)	(0.206)	(0.239)	(0.321)	(1.179)
First-stage F (S/L)	56 / 62	119 / 123	95 / 89	78 / 74	60 / 64
Observations (OLS / IV)	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077

Notes: Each column reports the column 3 specification of Table 1 at a different concentration cutoff p. The dependent variable is $\log(1 + \text{Patents}_{jt,g})$. Controls match Table 1, column 3, with Conc^p and RevConc^p replaced at the corresponding percentile. Standard errors, clustered at the 3-digit industry level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.2. *Breakthrough patents*

I next replace the dependent variable with $\log(1 + \text{BT}_{jt,g})$, where BT counts breakthrough patents, the top decile of the inventiveness measure of Appendix E. If the crowding-out result came from strategic filing, the significance of the results would be weakened for these patents.

Table F.2 reruns the four columns of Table 1 with this dependent variable. The small-firm coefficient is negative and significant in OLS in all four columns. Under IV it remains negative, though smaller in magnitude than in Table 1 and less precisely estimated. The large-firm coefficient is small and not significantly different from zero in columns 2–4 under both estimators.

TABLE F.2
Breakthrough patents as the dependent variable

	(1)	(2)	(3)	(4)
Panel A: OLS				
$\text{Conc}^{90} \times$ Small	-2.021^{***}	-0.542^{***}	-0.546^{***}	-0.371^{***}
	(0.119)	(0.137)	(0.137)	(0.075)
$\text{Conc}^{90} \times$ Large	2.387^{***}	0.258	0.265	0.123
	(0.218)	(0.262)	(0.256)	(0.078)
Panel B: Bartik IV				
$\text{Conc}^{90} \times$ Small	-2.214^{***}	-0.498^{*}	-0.506^{*}	-0.312^{**}
	(0.170)	(0.261)	(0.263)	(0.142)
$\text{Conc}^{90} \times$ Large	2.264^{***}	-0.257	-0.316	-0.144
	(0.183)	(0.257)	(0.266)	(0.161)
First-stage F (S/L)	7,374 / 6,972	95 / 92	95 / 89	195 / 175
Observations (OLS / IV)	5,664 / 4,077	5,664 / 4,077	5,664 / 4,077	13,431 / 9,900

Notes: Same specification as Table 1, with the dependent variable replaced by $\log(1 + \text{BT}_{jt,g})$, where BT is the count of breakthrough patents (top-decile inventiveness, Appendix E). Standard errors, clustered at the 3-digit industry level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.3. *Two-way clustering*

The baseline specification clusters standard errors by 3-digit industry. If the residuals are also correlated across industries within a year, for example, because of common shocks to subsidy allocation, these standard errors would be understated. Table F.3 reruns Table 1 with two-way clustering by industry and year.

The point estimates are the same as in Table 1. The IV standard errors in columns 2–4 are larger, but the small-firm coefficient remains significant at the 1% level.

F.4. *Sample splits*

Table F.4 reruns the column 3 specification on four restricted samples. I drop, in turn, the top 1% of firms by employment, the top 1% by subsidy receipts, firms whose reported subsidies exceed their revenue, and firms designated as technology SMEs. The first two samples remove the most extreme firms, the third removes likely cases of misreporting, and the fourth removes firms that receive support through a separate program.

The small-firm coefficient is negative and significant in every sample, between -0.65 and -1.19 in OLS and between -1.29 and -1.70 under IV. Dropping the largest subsidy recipients gives the smallest IV estimate in absolute value, -1.29, and the estimates in the other restricted samples are close to the baseline.

TABLE F.3
Two-way clustering by industry and year

	(1)	(2)	(3)	(4)
Panel A: OLS				
$\text{Conc}^{90} \times$ Small	-3.584^{***}	-0.675^{***}	-0.682^{***}	-0.557^{***}
	(0.182)	(0.164)	(0.153)	(0.098)
$\text{Conc}^{90} \times$ Large	2.802^{***}	0.242	0.240	0.215^{*}
	(0.121)	(0.219)	(0.208)	(0.101)
Panel B: Bartik IV				
$\text{Conc}^{90} \times$ Small	-4.168^{***}	-1.547^{***}	-1.559^{***}	-1.093^{***}
	(0.233)	(0.404)	(0.388)	(0.249)
$\text{Conc}^{90} \times$ Large	3.350^{***}	0.376	0.283	0.271
	(0.187)	(0.221)	(0.222)	(0.207)

Notes: Same specification as Table 1, with standard errors using two-way clustering at the 3-digit industry and year level (Cameron et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE F.4
Sample splits for the column 3 specification

	OLS		Bartik IV	
Sample	Small	Large	Small	Large
Baseline (Tab. 1, col 3)	-0.682^{***}	0.240	-1.559^{***}	0.283
	(0.171)	(0.189)	(0.343)	(0.239)
Drop top-1% emp.	-0.792^{***}	0.262	-1.704^{***}	0.536^{**}
	(0.172)	(0.164)	(0.387)	(0.261)
Drop top-1% subsidy	-1.194^{***}	-0.081	-1.287^{***}	0.326
	(0.223)	(0.222)	(0.255)	(0.201)
Drop subsidy > revenue	-0.655^{***}	0.276	-1.572^{***}	0.316
	(0.172)	(0.199)	(0.350)	(0.253)
Drop tech-SME	-0.678^{***}	0.128	-1.409^{***}	0.258
	(0.172)	(0.186)	(0.353)	(0.263)

Notes: Each row applies a firm-level filter to the working sample, then re-runs the column 3 specification of Table 1. Standard errors clustered at the 3-digit industry level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.5. *Pre-trend test*

The Bartik instrument B_{jt} uses persistent differences across provinces in their bias toward large firms, weighted by the 2008 provincial employment shares of each industry. If these shares were correlated with earlier trends in patenting that differ by size group, the instrument would be invalid. Following Jaeger et al. (2018), I regress the change in patenting within a cell, $\Delta y_{jt,g} = y_{jt,g} - y_{j,t-2,g}$, on $B_{jt} \times$ Small and $B_{jt} \times$ Large.

The coefficient on the small-firm interaction is 0.091 ($p = 0.12$), not significantly different from zero, so the small-firm estimate in Table 1 does not reflect earlier differ-

ences in patenting trends across industries. Meanwhile, the coefficient on the large-firm interaction is 0.155 ($p < 0.01$). It is small relative to the second-stage IV estimate but significant, so the large-firm IV estimate should be read with caution.

G. STANDARD ERRORS

This appendix describes the standard errors of the SMM estimates in Table 2.

Sandwich estimator. I use the SMM variance estimator of Gouriéroux and Monfort (1996), as in Garcia-Macia et al. (2019),

$$\widehat{\mathrm{Var}}(\hat{\theta}) \;=\; (G'WG)^{-1}\, G'W\, \hat{\Omega}\, WG\, (G'WG)^{-1}, \tag{G.1}$$

where $G \in \mathbb{R}^{7\times 6}$ is the Jacobian of the model moments with respect to the parameters at $\hat{\theta}$, $W \in \mathbb{R}^{7\times 7}$ is the weighting matrix in (24), and $\hat{\Omega} \in \mathbb{R}^{7\times 7}$ is the sampling covariance of the data moments. Since the model moments are computed without simulation, the simulation-noise term of the general formula is zero.

Bootstrap of $\hat{\Omega}$. I estimate the covariance of the moments with a firm-cluster bootstrap of the Chinese Tax Survey microdata. In each of $B = 500$ replicates, I draw $n = 52{,}091$ firms with replacement from the working sample and keep every observation of each drawn firm, so a firm drawn twice appears twice with its full history. I then recompute the inverse-probability weights that match the firm-size distribution of the Economic Census, using the bin shares of the bootstrap sample, and recompute the moments as in Section 3. Resampling firms preserves the serial correlation within firms on which the slope moments depend. For each moment, $\hat{\Omega}$ uses the variance of the estimator that produces the corresponding data target, and (G.1) carries this uncertainty to the six parameters.

Numerical Jacobian. I compute G by central differences in the logs of the parameters,

$$(G_{\log})_{ik} \;=\; \left.\frac{\partial m_i}{\partial \log \hat{\theta}_k}\right|_{\hat{\theta}} \;\approx\; \frac{m_i(\hat{\theta}\cdot e^{+h}e_k) - m_i(\hat{\theta}\cdot e^{-h}e_k)}{2h}, \tag{G.2}$$

with $h = 10^{-2}$, a one percent perturbation of each parameter. Differentiating in logs allows me to apply the same relative step to parameters of different magnitudes while keeping every perturbed parameter positive. It also multiplies each column of the raw Jacobian by the corresponding parameter, $G_{\log} = G_{\mathrm{raw}}\cdot\mathrm{diag}(\hat{\theta})$, which puts the columns on a comparable scale and improves the conditioning of $G'WG$.

Reporting scale. Table 2 reports standard errors on the original scale of each parameter. Since $G_{\log}$ is the Jacobian with respect to the logs of the parameters, they follow from (G.1) by the delta method, $\mathrm{SE}(\hat{\theta}_k) = \hat{\theta}_k\, \mathrm{SE}(\log \hat{\theta}_k)$.

H. SUPPLEMENTARY ESTIMATION RESULTS

H.1. *The Pat/R&D profile*

Figure H.1 plots the size profile of Pat/R&D in the model as the cost scale c_ψ varies, with the other parameters held at their estimates. At $\hat{c}_\psi = 0.32$, the profile falls from 0.24 at $n = 1$ to 0.04 at $n = 15$. As c_ψ rises the profile flattens, and in the Klette–Kortum limit it is constant in n, as the argument of Section 4.3 implies.

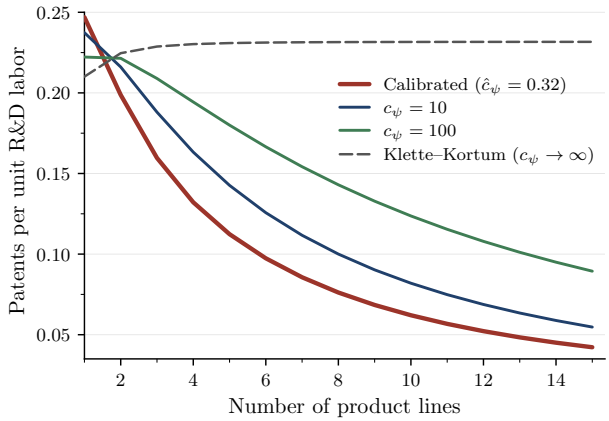


FIGURE H.1
Patents per unit of R&D labor as c_ψ varies

Notes: The figure plots the model-implied schedule $\text{Pat}(n)/\text{R\&D}(n)$ against the number of product lines n at four values of the process-efficiency cost scale c_ψ: 0.03, the estimated $\hat{c}_\psi = 0.32$, 30, and the Klette–Kortum limit $c_\psi \to \infty$. The remaining parameters $(\omega, \theta, \lambda, H, \alpha_e)$ are held at their estimated values.

H.2. *Re-estimation without process efficiency*

To shut down the process efficiency mechanism, I re-estimate the model with $c_\psi \to \infty$, minimizing the SMM objective over $(\theta, \lambda, H, \alpha_e)$. The fitted slope of log(Pat/R&D) is $+0.12$ rather than -0.56, and the other three moments in Table H.1 are further from the data than under the full model.

TABLE H.1
Model fit: full versus restricted model

Moment	Data	Full model	Restricted
Aggregate growth	0.0222	0.0231	0.0407
IQR(log Emp)	1.39	1.08	0.60
log(R&D/Rev) slope	−0.61	−0.52	−1.24
log(Pat/R&D) slope	−0.56	−0.55	+0.12

Notes: The restricted model shuts down process efficiency by imposing $c_\psi \to \infty$ and re-estimates the remaining parameters against the benchmark SMM objective. The restricted estimates are $\hat{\theta} = 0.36$, $\hat{\lambda} = 0.55$, $\hat{H} = 1.00$, and $\hat{\alpha}_e = 2.78$. Slopes are coefficients from regressions of the indicated log ratio on log employment.

H.3. *The crowding-out coefficient in the model*

As a check on the general-equilibrium channels of the model, I ask whether the calibrated model reproduces the crowding-out coefficient of the preferred specification in Table 1. With all parameters at their estimates, I impose nineteen subsidy schedules that raise the rate at the top of the size distribution while holding the small-firm rate fixed, and I solve for the equilibrium under each. I then regress the gap in log patents per firm between small and medium firms, $\log(\text{Pat}_S^{\text{avg}}) - \log(\text{Pat}_M^{\text{avg}})$, on Conc^{90} across the nineteen equilibria, treating the model as a single industry-year.[4] The small-firm coefficient in the model is -0.38 ($R^2 = 0.97$), slightly over half of the OLS estimate in column 3 (-0.68). Two features of the model may account for the gap. First, its firm-size distribution is more compressed than in the data, which dampens the responses of the skilled wage and of blocking to subsidy concentration. Second, the model omits other channels through which subsidy concentration can hurt small firms, such as credit constraints, which bind hardest on small firms.

H.4. *The calibrated economy*

Figure H.2 plots two features of the economy using the calibrated parameters in Table 3 of the main text. Panel (a) shows that the share of R&D labor devoted to process efficiency rises from 13% at single-line firms to 67% at firms with eleven lines. Panel (b) shows that the probability that an attempt against a firm with n lines fails rises from close to zero for single-line firms to above 0.12 for firms with eleven or more lines.

H.5. *Consumption-equivalent variation*

I evaluate welfare on the balanced growth path with the household's CRRA preferences and initial consumption net of the tax, $C_0 = \hat{Y} - T$. Let (C_0, g_0) denote baseline consumption and growth, and (C_1, g_1) consumption and growth under the alternative policy. The CEV ξ solves $W(\xi C_0, g_0) = W(C_1, g_1)$, so that

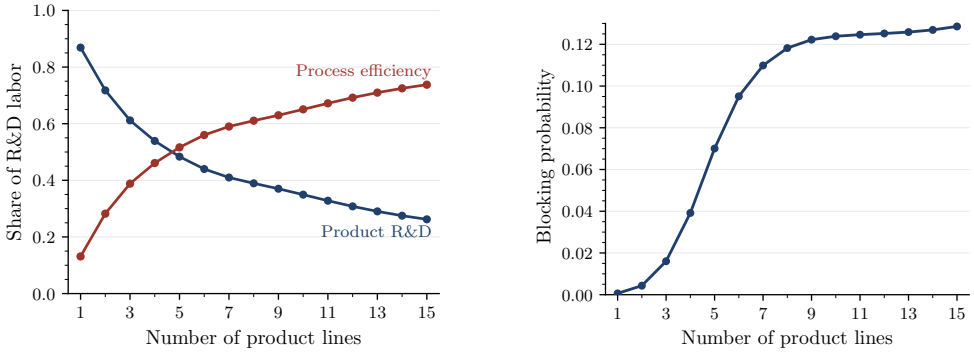
$$\xi = \frac{C_1}{C_0} \left[\frac{\rho - (1-\vartheta)\, g_0}{\rho - (1-\vartheta)\, g_1} \right]^{1/(1-\vartheta)}. \tag{H.1}$$

I report $\xi - 1$ in percent.

I. COMPUTATIONAL ALGORITHM

The algorithm has two nested loops. The inner loop takes the skilled wage w_s as given and computes the remaining objects of the balanced growth path. The outer loop

4. The baseline gives all firms a uniform 5% discount on the cost of R&D labor. Each step then adds 2 percentage points to the discount of firms above p99 and 1 point to that of firms in p90–p99, roughly in line with the data, and leaves the other firms unchanged. The nineteen schedules generate values of Conc^{90} between 0.28 and 0.80. Small firms are those below p47 of employment and medium firms those between p47 and p86, which match the 35- and 395-employee cutoffs in the data. The dependent variable is $X^* B$, the rate of successful product innovations per firm, which is the model analog of patents per firm.



(a) R&D composition. (b) Blocking probability.

FIGURE H.2
Features of the calibrated economy

Notes: Panel (a) plots the composition of R&D labor against the number of product lines n. Panel (b) plots the probability that an attempted innovation against a size-n incumbent fails. Both panels are computed using the calibrated parameters in Table 3 of the main text.

bisects on w_s to clear the market for skilled labor (18).

The inner loop proceeds as follows.

1. Initialize the aggregate attempt rate $\bar{\tau}$, the share of active lines Φ_A, average process efficiency $\bar{\Psi}$, and the distribution of firms $\mu(n)$ on a grid $n \in \{1, \ldots, N_{\max}\}$.

2. Compute the profit coefficient $\tilde{p}$ from (B.2).

3. Compute process efficiency $\psi^*(n)$ from (5). Under a subsidy schedule, map firm sizes to employment, assign the subsidy rates $\sigma(n)$, and use the effective wage $(1 - \sigma(n))w_s$ in place of w_s.

4. Compute the success probabilities $B(n)$ and displacement rates $\tau(n)$ from (11) and (12).

5. Solve for the value function $\hat{V}(n)$ by policy iteration, using the Thomas algorithm for the resulting tridiagonal system.

6. Obtain the attempt rates $X^*(n)$ from the first-order condition (15).

7. Compute the entry rate X_e from the free-entry condition (16), by bisection when $\alpha_e > 0$ and as residual labor when $\alpha_e = 0$.

8. Update $\mu(n)$ by solving the flow equations for the number of lines, again a tridiagonal system, with the entry source at $n = 1$.

9. Update Φ_A, $\bar{\Psi}$, $\bar{\tau}$, and the mass of firms $F = \Phi_A/\bar{n}$.

10. If μ has not converged, set $\mu \leftarrow (1 - \delta)\mu + \delta\mu_{\text{new}}$ with $\delta = 0.4$ and return to step 2.

To save time, I set $N_{\max} = 80$ and raise it to $N_{\max} = 640$ only when more than 0.5% of the mass of μ lies above $0.85N_{\max}$. The convergence tolerance is 10^{-7} in the inner loop and 10^{-6} in the outer bisection.

J. ROBUSTNESS OF THE QUANTITATIVE RESULTS

This appendix reports the parameter estimates and welfare gains for the robustness exercises of Section 5.4.

J.1. *Parameter estimates*

Table J.1 reports the estimates under the five specifications. All of them stay within about one standard error of the benchmark. The entry-cost parameters H and α_e move the most, since they partly substitute for each other in the free-entry condition.

TABLE J.1
Parameter estimates across robustness specifications

Parameter	Description	Benchmark	+ Yield	+ Process	$\gamma = 0.4$	$\gamma = 0.6$
ω	Process efficiency cost convexity	3.279	3.202	3.223	3.276	3.198
c_ψ	Process efficiency cost scale	0.323	0.260	0.339	0.361	0.398
θ	Incumbent innovation capacity	0.149	0.145	0.149	0.149	0.166
λ	Quality step size	1.079	1.177	1.072	1.047	1.005
α_e	Entry-cost convexity	3.000	2.615	3.000	2.914	3.000
H	Entry-cost level	0.527	0.328	0.534	0.476	0.598

Notes: Each column re-estimates the model under a different specification. "+ Yield" keeps the seven benchmark moments and adds an eighth targeted moment, the ratio of patents per subsidy yuan for subsidized firms in the p50–p90 size bin relative to firms below p50, computed within 4-digit industry × year cells. "+ Process" instead adds the share of large product innovators that also conduct process innovation (87.1%) as the eighth moment (Section 5.4). The γ columns fix the R&D production-function curvature and re-estimate the remaining six parameters against the benchmark moments. The externally calibrated parameters remain unchanged.

J.2. *Welfare outcomes*

Table J.2 reports the welfare gains under each specification for the three counterfactuals of Section 5. The conclusions do not change. Removing the existing subsidies raises welfare by about 1% in consumption-equivalent terms, the budget-neutral optimum by 10–14%, and the variable-budget optimum by 18–22%.

The variable-budget gain varies the most, from 18.0% to 22.4%, because this counterfactual is the most sensitive to the quality step λ and to the curvature γ of the R&D production function.

K. FIRM TYPE HETEROGENEITY EXTENSION

In the main text, firms are ex ante identical, and size is the only thing that distinguishes between two firms. However, large firms may be large because they are better at innovating, as in Acemoglu et al. (2018). If that is the case, subsidizing them may

TABLE J.2
Welfare gains across robustness specifications (CEV $\xi - 1$, %)

Counterfactual	Benchmark	+ Yield	+ Process	$\gamma = 0.4$	$\gamma = 0.6$
No subsidy	1.02	1.11	1.03	1.08	0.97
Budget-neutral optimum	10.73	12.33	10.69	13.64	12.59
Variable-budget optimum	18.20	21.54	18.04	21.39	22.40

Notes: Each entry is the consumption-equivalent welfare gain (CEV $\xi - 1$ in percent) relative to the baseline calibration of the corresponding specification, computed at $\rho = 0.020$ and $\vartheta = 2$. "+ Yield" adds the ratio of patents per subsidy yuan for subsidized firms in the p50–p90 size bin relative to firms below p50 as an eighth targeted moment, and "+ Process" adds the share of large product innovators that also conduct process innovation (Section 5.4). The γ columns fix the R&D production-function curvature and re-estimate the remaining parameters. The benchmark column reproduces Panel D of Table 5.

be less costly than the baseline suggests. In this appendix I allow firms to differ permanently in their innovation productivity.

Adding firm types does not change how process efficiency is chosen or how contests are decided. The process efficiency in (5) does not depend on a firm's type, and the outcome of a contest depends only on the process efficiencies of the two firms, so two firms with the same portfolio block the same set of challengers regardless of type. Instead, types change how often firms attempt and, in turn, how firms are distributed over portfolios: high types grow faster and make up a larger share of large firms.

A subsidy to large firms therefore applies to more high-type firms, but at those firms, too, it funds mostly process efficiency, so the blocking externality remains. Which effect dominates is a quantitative question that I leave to a re-estimation of the model with types. In any case, a planner who could condition on type as well as size would do at least as well as the size-based planner of Section 5, since the size-based schedules remain available to it.

K.1. *Setup*

When an entry attempt succeeds, the new firm draws a permanent type θ_f. With probability α it is a high type, $\theta_f = \theta^h$, and otherwise a low type, $\theta_f = \theta^l < \theta^h$. A firm of type θ_f generates attempts at the rate

$$X = \theta_f \, h_X^{1-\gamma},$$

in place of (8). The rest of the model is unchanged.

K.2. *Process efficiency and blocking*

The process-efficiency problem (4) does not involve θ. Hence the optimal choice is

$$\psi^*(Q) = \left(\frac{\tilde{p} \, Q}{\omega \, w_s \, c_\psi} \right)^{1/(\omega - 1)},$$

as in (5), and the flow profit $\pi(Q) = \tilde{p}\,Q + \hat{D}\,Q^{\eta}$ and process R&D labor $h_{\psi} = c_{\psi}[\psi^{*}(Q)]^{\omega}$ depend on a firm's portfolio only through its aggregate quality.

Nor do types affect blocking, since the advantage $A(\psi, \psi') = (1+\lambda)(1+\psi)-(1+\psi')$ depends on the process efficiencies of the challenger and the incumbent but not on their types. The success probability keeps the form of (11), and the displacement rate (12) becomes

$$\tau(\psi) = X_e\,\Lambda(A(0,\psi)) + \sum_{k\in\{h,l\}} \int X^{k}(p'')\,\Lambda(A(\psi(p''),\psi))\,dM_k(p''),$$

where M_k is the measure of type-k firms over portfolios. Proposition 1 holds because its proof requires only that ψ^{*} be increasing in Q, and Proposition 2 holds within each type.

K.3. *Innovation and the distribution of firms*

Types affect the firm's dynamic problem only through the cost of product R&D. Let $\hat{V}_k(p)$ denote the value of a type-k firm with portfolio p, and $G_k(p) \equiv \mathcal{G}\hat{V}_k(p)$ its expected gain from an attempt, with $\mathcal{G}$ given by (13). The value function (14) then becomes

$$\hat{r}\,\hat{V}_k(p) = \max_{X\geq 0}\left\{\pi(Q(p)) - w_s\left[\frac{X}{\theta_k}\right]^{\frac{1}{1-\gamma}} + X\,G_k(p)\right\} - \tau(\psi(p))\,\mathcal{L}\hat{V}_k(p) - g\sum_j x_j\,\frac{\partial \hat{V}_k(p)}{\partial x_j}$$

for $k \in \{h,l\}$, with the same discount rate $\hat{r} = \rho + (\vartheta - 1)g + \varphi$ as in the baseline. The first-order condition gives

$$X^{k}(p) = \theta_k^{1/\gamma}\left[\frac{(1-\gamma)\,G_k(p)}{w_s}\right]^{(1-\gamma)/\gamma}.$$

Since the exponent on θ_k is $1/\gamma > 1$, a small difference in type becomes a larger difference in the attempt rate. Moreover, $\hat{V}_h(p) \geq \hat{V}_l(p)$ at every portfolio, since a high type can follow the policy of a low type at a lower R&D cost.

Since a firm keeps its type forever, high- and low-type firms form two separate populations, and the stationary distribution can be described by one measure for each type. Each M_k satisfies (17) with X^{k} in place of X^{*} and with the entry term multiplied by α_k, where $\alpha_h = \alpha$ and $\alpha_l = 1-\alpha$. High types attempt more at every portfolio, so they grow faster and are overrepresented among firms with high aggregate quality.

K.4. *Equilibrium*

Since a firm learns its type only after its entry attempt succeeds, a potential entrant evaluates an attempt at the expected value of a new firm, $\alpha\,\hat{V}_h + (1-\alpha)\,\hat{V}_l$, and the free-entry condition (16) holds with $G(\emptyset)$ computed from this average. The other equilibrium conditions require only small changes. In the market for skilled labor (18), product R&D is now summed over the two types, while process R&D still depends only

on aggregate quality. The aggregate attempt rate adds the attempts of both types of incumbents to those of entrants, and the growth rate keeps the form of (19), with S now induced by $M_h + M_l$. Because blocking depends only on process efficiency, the success probabilities and displacement rates are the same functions of ψ as in the main text. Finally, welfare and the consumption-equivalent variation (H.1) are computed exactly as before.

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